Clinical Nutrition ESPEN 45 (2021) 267–274

Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: http://www.clinicalnutritionespen.com

Original article

Precision and accuracy of bioelectrical impedance analysis devices in supine versus standing position with or without retractable handle in Caucasian subjects

Katrin Hamilton-James, Tinh-Hai Collet, Claude Pichard, Laurence Genton, Yves M. Dupertuis*

Clinical Nutrition, Geneva University Hospitals, Rue Gabrielle-Perret-Gentil 4, 1211, Geneva 14, Switzerland

ARTICLE INFO

Article history:
Received 8 August 2021
Accepted 12 August 2021

Keywords:
Bioelectrical impedance analysis
Body composition
Dual-energy X-ray absorptiometry
Fat mass
Nutritional assessment

SUMMARY

Background & aims: Bioelectrical impedance analysis (BIA) could be facilitated in subjects who are able to stand by using scales without (BIA_{std4}) or with a retractable handle (BIA_{std8}), provided that they are as precise as BIA devices commonly used in the supine position in the hospital setting (BIA_{sup}). This observational prospective cross-sectional study aimed to compare the precision and accuracy of BIA_{std4}, BIA_{std8} and BIA_{sup} in a Caucasian population.

Methods: Fat mass percentage (FM%) was measured in 160 healthy Caucasian subjects (80 men/80 women) aged 20–60 years, with a body mass index (BMI) $\geq$18.5 and < 30 kg/m^2, using the HAGRID Body Fat Scales (Huawei Technologies Co., Ltd., China) in BIA_{std4} or BIA_{std8} mode, and the Nutriguard-M (Data Input GmbH, Germany) as BIA_{sup}. Intra-unit and inter-unit precisions of each device were evaluated by calculating the coefficients of variation (CV%) of 3 measurements with 3 different units of each device. Inter-device precisions were evaluated with Pearson correlations, Bland–Altman plots, and repeated measures ANOVA followed by post-hoc Bonferroni tests. Accuracy of these BIA devices was estimated in a subgroup of 16 subjects, using comparison with dual-energy X-ray absorptiometry (DXA).

Results: The study population was 40 ± 12 years old, with a body height and weight of 171 ± 10 cm and 72.2 ± 11.5 Kg, respectively. All three devices were very precise with intra-unit CV% of 0.5%, 0.9%, and 0.3% and inter-unit CV% of 0.5%, 1.1%, and 0.4% for BIA_{std4}, BIA_{std8} and BIA_{sup}, respectively. Inter-device precision was ±2.1% for BIA_{std4}/BIA_{sup}, ±1.9% for BIA_{std8}/BIA_{sup}, and ±1.3% for BIA_{std8}/BIA_{std4}. Bland–Altman plots showed bias ±1.96 SD of 0.3 ± 5.2% for BIA_{std4}/BIA_{sup}, −0.4 ± 4.5% for BIA_{std8}/BIA_{sup} and −0.6 ± 3.1% for BIA_{std8}/BIA_{std4}. Compared to DXA, all three devices tended to underestimate FM% in men with low BMI, while only BIA_{std4} and BIA_{std8} tended to overestimate FM% in women with high BMI. FM% measurement accuracy was ±2.6% for BIA_{sup}/DXA, ±3.3% for BIA_{std4}/DXA, and ±3.4% for BIA_{std8}/DXA.

Conclusions: Both BIA_{std4} and BIA_{std8} show a good intra- and inter-unit precision close to BIA_{sup}, making them suitable for rapid body composition assessment in non-bedridden subjects. However, all these three devices should not be used interchangeably, because BIA_{std4} and BIA_{std8} tend to accentuate FM% changes during body composition monitoring compared to BIA_{sup} and DXA.

Trial registration: ClinicalTrial.gov no. NCT04504799.

© 2021 The Author(s). Published by Elsevier Ltd on behalf of European Society for Clinical Nutrition and Metabolism. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Abbreviations: BIA, bioelectrical impedance analysis; BMI, body mass index; BW, body weight; CV%, coefficient of variation; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass; FM%, percent fat mass; LoA, limits of agreement; R, resistance; Xc, reactance.

* Corresponding author. Fax: + 41 22 372 93 63.

E-mail addresses: Katrin.Hamilton-James@hcuge.ch (K. Hamilton-James), Tinh-Hai.Collet@hcuge.ch (T.-H. Collet), Claude.Pichard@hcuge.ch (C. Pichard), Laurence.Genton@hcuge.ch (L. Genton), yves.m.dupertuis@hcuge.ch (Y.M. Dupertuis).

https://doi.org/10.1016/j.clnesp.2021.08.010
2405-4577/© 2021 The Author(s). Published by Elsevier Ltd on behalf of European Society for Clinical Nutrition and Metabolism. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Body composition measurement is of great use not only to assess the physical condition of healthy people but also to monitor the nutritional status of patients during their treatment course [1,2]. The most accurate methods to measure body composition, such as neutron activation, isotope dilution, hydro-densitometry, plethysmography, magnetic resonance, or X-ray imaging are complex, expensive and time-consuming. Some are even potentially harmful due to radiation exposure [3]. Only non-invasive indirect methods, such as bioelectrical impedance analysis (BIA), are suitable for routine clinical monitoring [4]. BIA measures the resistance (R) and reactance (Xc), or the impedance (Z) and phase angle (ϕ), when an alternating current of low intensity ($\leq$1 mA) and high frequency (5–1000 kHz) flows between two electrodes placed at the body extremities. R is therefore inversely proportional to the quantity of water and electrolytes in the body while Xc represents the capacity of the body cell membranes to act as mini-condensers against the current frequency. Prediction equations then make it possible to calculate the fat-free mass (FFM) and, by deduction, the fat mass (FM) from measured R, Xc, Z, ϕ and anthropometric data (i.e. body weight (BW), height and sex) [5]. As the current is imperceptible, BIA does not cause any discomfort for the subject. BIA is therefore a safe, cost-effective, quick and easy-to-use method for assessing body composition [6]. However, BIA measurements are affected by over or underhydration, the prediction equation, electrode placement, and device used [7]. In the past, our team demonstrated that the results could differ greatly from one prediction equation to another [8]. For this reason we developed our own prediction equation from a double cross validation with dual-energy X-ray absorptiometry (DXA) [9], and have utilized it for almost 20 years in a large number of publications in hospitalized patients as well as in healthy subjects [10–12]. The placement of the electrodes on the body plays an important role as well [13]. Bipolar BIA devices with foot-to-foot or arm-to-arm arrangement (4 electrodes) are believed to be less reliable than tetrapolar BIA devices with foot-to-foot plus hand-to-hand arrangements (8 electrodes), where the current flows through the whole body [14–16]. In the hospital setting, BIA is usually performed in the supine position with a hand-to-foot arrangement (4 electrodes) on the same body side [17]. This method measures the current flowing through the hemibody of bedridden patients. However, BIA measurement in the supine position is time-consuming because it requires the subject to lie long enough for the blood volume to spread evenly throughout the body. It also requires an operator, who applies single-use electrodes and reports the raw data into a software calculating FFM and FM. Newly released BIA scales with new algorithms and tetrapolar technologies could therefore facilitate body composition assessment in subjects who are able to stand and without the requirement of another operator. The fluid and electrolyte distribution in the body is however not the same when standing as when lying down [18], so the current does not flow in the same way and does not give the same R, Xc, Z and ϕ values [19]. This is why new prediction equations have to be developed for every new BIA device using different technology to give FFM and FM values as close as possible to the true values given by a criterion method.

The aim of this study was to evaluate the precision and accuracy of new BIA devices in the standing position without (BIA_{std4}) or with a retractable handle (BIA_{std8}) in a Caucasian population, compared to a BIA device in the supine position commonly used in clinical practice (BIA_{sup}).

2. Materials and methods

2.1. Study design and objectives

This observational prospective cross-sectional study was carried out at the Geneva University Hospitals with the approval of the Cantonal Research Ethics Committee (BASEC no. 2020–01613; ClinicalTrials.gov no. NCT04504799). The aim was to evaluate the precision and accuracy of FM% measurement with BIA_{std4} and BIA_{std8} in healthy Caucasian adults, in comparison with routinely used BIA_{sup} and DXA as criterion method [20].

2.2. Subjects

Since the average primary outcome (FM%) is 29 $\pm$ 8% in our local database, we estimated that the sample size should be at least 154 subjects to be 90% sure that the limits of a two-sided 90% confidence interval exclude a difference of more than 3% between the BIA devices, using an on-line power calculator for equivalence trial (https://evaluation.sealedenvelope.com/power/continuous-equivalence/). Consequently, a total of 160 healthy Caucasian subjects aged from 20 to 60 years, with BMI $\geq$ 18.5 and < 30 kg/m^2 were recruited from the local population through posters and social media advertising, using a stratification sampling method based on age, sex and BMI to be as representative as possible to the general population (Table S1). Exclusion criteria were the inability to understand the study procedures, hospitalization in the previous 6 months, physical disability, medication impacting fluid and electrolyte balance (e.g. diuretics or calcium channel blockers), moderate to intense physical activity 2–3 h before BIA measurement and pregnancy.

2.3. Procedure

Participants signed an informed consent form and all clinical data were recorded in a secure coded form. BW and height were measured with minimal clothing and barefoot, using a calibrated weight scale and stadiometer, respectively. BIA measurements on each subject were then carried out 3 times utilizing 3 different units (A, B, C) of each BIA_{std4}, BIA_{std8} and BIA_{sup} according to the following sequence: A, B, C, B, C, A, C, A, B. Thus, each subject underwent 9 BIA measurements with each device. The repetition of measurements within a unit aimed to assess the intra-unit precision and the repetition of measurements between the 3 units was intended to assess the inter-unit precision of each device. This procedure was repeated with the three BIA devices one after the other for each participant to assess the inter-device precision. To minimize variations due to fluid shifts in the body [18], the different BIA units were placed side by side so that the subject could move from unit to unit without wasting time and too much movement, starting with the measurements in standing position and ending with the measurements in supine position. The latter was carried out by interchanging the BIA devices according to the sequence specified above, after the subject had been lying down for approximately 5 min, according to the manufacturer's instructions. The aim of this study being the comparison of the intra- and inter-unit precision of the BIA devices, we kept the same adhesive electrodes for all the measurements in supine position to reduce time between measurement and avoid error due to electrode replacement. The whole procedure lasted about 30 min.

2.4. Bioelectrical impedance analysis in the standing position

We evaluated the new HAGRID Body Fat Scale HAG-B19 (Huawei Technologies Co., Ltd., Shenzhen, China), which has just been released and can be used without (BIA_{std4}) or with a retractable handle (BIA_{std8}). This phase-sensitive BIA scale calculates FFM and thus FM% in both modes from the undisclosed Trufit™ algorithm, after applying an alternative current of 0.8 mA at 50 and 250 kHz through the body. For the measurement, height, age and sex of each participant were entered into the smartphone Huawei Fat Scale application connected via Bluetooth to the scale. The subjects were asked to turn off and remove their smartwatches to avoid any interferences. For BIA_{std4}, subjects stood barefoot on the scale with their feet positioned on the 4 integrated electrodes. For BIA_{std8}, subjects were additionally asked to take the retractable handle and hold it firmly with their arms extended at a 30° angle to the trunk. The retractable handle is equipped with 4 additional electrodes, thus allowing the current to flow from the feet to the hands, as it is the case in BIA_{sup}. BW, R and Xc were measured simultaneously by a strain-type pressure sensor and 4 reusable stainless plate electrodes (BIA_{std4}) or 8 electrodes (BIA_{std8}), respectively. Generated data was sent via Bluetooth to the smartphone and processed on the Huawei Health application.

2.5. Bioelectrical impedance analysis in the supine position

For the BIA_{sup} comparator, we used the Nutriguard-M (Data Input GmbH, Pöcking, Germany) according to our aforementioned clinical routine [21]. Subjects were asked to remove all metal objects (e.g. watches or jewelry) prior to lying supine and barefoot on a consultation bed, with limbs positioned slightly away from the body. After cleaning the skin with a hydroalcoholic solution, two self-adhesive disposable electrodes (3M® Red Dot®, 3M Health Care, Neuss, Germany) were placed on the right hand and wrist, and two on the right foot and ankle [1]. This phase-sensitive multifrequency device applies an alternating electrical current of 0.8 mA at 5, 50 and 100 kHz to these electrodes. The measured R and Xc values at 50 kHz were used to calculate the FFM and FM% from our previously validated prediction equation [9]:

$$FFM = -4.104 + (0.518 \times height^2/R) + (0.231 \times BW) + (0.130 \times Xc) + (4.229 \times sex;\ men = 1,\ women = 0) \Rightarrow FM\% = (BW - FFM) \times 100/BW.$$

2.6. Dual-energy X-ray absorptiometry

Accuracy of FM% measurement with the different BIA devices was further evaluated in a subgroup of 16 subjects using whole-body DXA (Horizon A, Hologic®, Inc., Bedford, USA). Each of the 16 subjects represented one of the 16 categories defined in Table S1. DXA measurement was carried out on the same day just prior to the BIA measurements. The subject laid supine within the border defined on the scanning table while the X-ray source below and the detector above scanned the whole-body from front to back. The system utilized a constant X-ray source producing fan beam dual-energy radiation with effective dose equivalents of 2–5 μSv.

FM% was estimated based on the tissue attenuation of two different X-ray energies and calculated as:

$$FM\% = FM/(FM + \text{bone-free lean tissue mass} + \text{bone mineral content}) \times 100.$$

2.7. Data handling and statistical analysis

All BIA measurements were carried out in triplicates with each of the three units of the three devices on each of the 160 subjects. All variables were expressed as proportions or means ± 1 standard deviation (SD), unless specified. Repeated measures ANOVA followed by post-hoc Bonferroni tests were performed to highlight statistical differences between the different sex and BMI categories. To evaluate the intra-unit precision, the coefficient of variation (CV % = SD/mean*100) was calculated from the mean values ± SD of FM % measured by each unit. To evaluate the inter-unit precision, CV% was calculated from the mean ± SD of the three mean values of FM% measured by each of the three units. Results were compared with Pearson correlations and paired *t*-tests. Bland–Altman plots were performed to assess the linear relationship, possible systematic errors, and limits of agreement (LoA) between the different BIA devices. Inter-device precision was calculated as the average of the absolute values of the differences between the devices, while accuracy was calculated as the average of the absolute values of the differences between the BIA devices and DXA. Statistical analyses were performed with the Stata/IC 13.1 software for Windows (StataCorp LP, College Station, TX, USA). Statistical significance level was set at $P < 0.05$.

3. Results

3.1. Population description

The study population was 40 ± 12 years old, height of 171 ± 10 cm, and BW of 72.2 ± 11.5 Kg. Women were on average 13 cm shorter and weighed 11.9 Kg less than men (*t*-test, $P < 0.001$). Overweight subjects were on average 14.3 Kg heavier and BMI of 5.3 Kg/m^2 more than normal weight subjects (*t*-test $P < 0.001$). The different categories by BMI and sex are detailed in Table 1.

3.2. Intra- and inter-unit precision of FM% measurement

The difference of FM% measurements within and between the three units of each device was very low (Fig. 1). Comparisons of the three measurements within the same unit indicated a high intra-unit precision for all three devices with a CV% < 1% (Table 2). Comparison of measurements between the three units showed also a high inter-unit precision for all three devices with a CV% ≤ 1.1% (Table 3).

3.3. Inter-device precision of FM% measurement

Correlations of FM% measurement between the different devices were very strong with a Pearson coefficient rho of 0.95 for BIA_{std4}/BIA_{sup}, 0.97 for BIA_{std8}/BIA_{sup}, and 0.98 for BIA_{std8}/BIA_{std4} (Fig. 2). Bland–Altman plots showed LoA with bias ±1.96 SD of 0.3 ± 5.2% for BIA_{std4}/BIA_{sup}, −0.4 ± 4.5% for BIA_{std8}/BIA_{sup} and −0.6 ± 3.1% for BIA_{std8}/BIA_{std4} (Fig. 3). Inter-device precision was ±2.1% for BIA_{std4}/BIA_{sup}, ±1.9% for BIA_{std8}/BIA_{sup}, and ±1.3% for BIA_{std8}/BIA_{std4}.

3.4. FM% measurements according to sex and BMI

Compared to BIA_{sup}, BIA_{std4} and BIA_{std8} gave similar results in men. In contrast, both tended to underestimate FM% in women with low BMI and overestimate FM% in women with high BMI (Table 4). The under/overestimation cut-off for FM% measurement

Table 1
Physical characteristics of the participants according to categories of normal weight (N) or overweight (O) men or women.

	Both sexes			Men		Women	
	All	N	O	N	O	N	O
Number	160	80	80	40	40	40	40
Age (y)	40 ± 12	40 ± 12	40 ± 12	39 ± 12	39 ± 12	40 ± 12	40 ± 12
Height (cm)	171 ± 10	172 ± 9	171 ± 10	179 ± 6[c,d]	178 ± 8[c,d]	165 ± 6[c,d]	164 ± 7[c,d]
BW (Kg)	72.2 ± 11.5	65.0 ± 9.8[a]	79.3 ± 8.3[a]	72.2 ± 7.0[b,c]	84.0 ± 7.5[b,c,d]	57.8 ± 6.1[b,c,d]	74.7 ± 6.2[b,c]
BMI (Kg/m^2)	24.5 ± 3.1	21.8 ± 1.7[a]	27.2 ± 1.6[a]	22.5 ± 1.5[b,c,d]	26.6 ± 1.4[b,c,d]	21.2 ± 1.6[b,c,d]	27.8 ± 1.6[b,c,d]

[a] Significant between BMI categories (*t*-test $P < 0.001$).
[b] Significant between BMI categories of the same sex.
[c] Significant between sexes of the same BMI category.
[d] Significant between different sexes and BMI categories (one-way ANOVA post-hoc Bonferroni test $P < 0.05$).

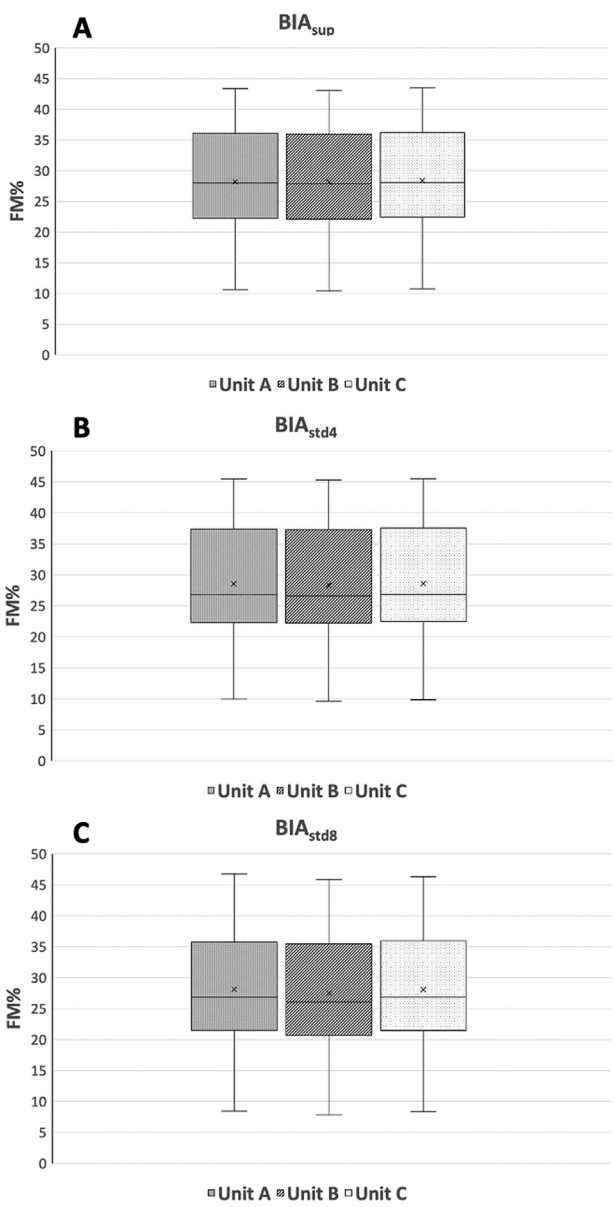

Fig. 1. Box plots showing distribution of FM% values (minimum, lower quartile, median, upper quartile, maximum) obtained by each of the three units of BIA_{sup} (A), BIA_{std4} (B), and BIA_{std8} (C).

Table 2
Intra-unit precision expressed as the coefficient of variation (CV %) within 3 measurements of FM% per unit (A, B, C) for each subject.

	A	B	C	Mean ± SD
BIA_{sup}	0.3	0.3	0.3	0.3 ± 0.0
BIA_{std4}	0.8	0.4	0.4	0.5 ± 0.2
BIA_{std8}	0.9	0.9	0.9	0.9 ± 0.0

Table 3
Inter-unit precision expressed as the coefficient of variation (CV %) of the FM% between the averages of 3 measurements per unit (A, B, C) for each subject.

	A	B	C	Mean ± SD	%CV
BIA_{sup}	28.2	28.1	28.4	28.2 ± 0.1	0.4
BIA_{std4}	28.5	28.3	28.6	28.5 ± 0.1	0.5
BIA_{std8}	28.1	27.5	28.0	27.9 ± 0.3	1.1

in women was at BMI of 23.0 Kg/m^2 for BIA_{std4}/BIA_{sup}, and 24.7 Kg/m^2 for BIA_{std8}/BIA_{sup}(Fig. 4).

3.5. Accuracy of FM% measurement compared to DXA

The accuracy of FM% measurements with these different BIA devices was assessed in a representative subgroup of 16 subjects using DXA. Each of the 16 subjects represented one of the 16 categories defined above, with an average age of 40 ± 12 years, BW of 72.3 ± 10.4 kg, height of 170.5 ± 10.7 cm and BMI of 25.0 ± 3.6. The results showed a good correlation with DXA for all three devices with a Pearson coefficient rho of 0.95 for BIA_{sup}/DXA, 0.91 for BIA_{std4}/DXA, and 0.93 for BIA_{std8}/DXA. FM% measurement accuracy was ±2.6% for BIA_{sup}/DXA, ±3.3% for BIA_{std4}/DXA, and ±3.4% for BIA_{std8}/DXA. All three BIA devices tended to underestimate FM% in men with low BMI, while only BIA_{std4} and BIA_{std8} tended to overestimate FM% in women with high BMI (Table 5). The under/overestimation cut-offwas at BMI of 28.5 Kg/m^2 and FM% of 38.4% for BIA_{sup}/DXA, BMI of 24.9 Kg/m^2 and FM% of 31.9% for BIA_{std4}/DXA, and BMI of 25.8 Kg/m^2 and FM% of 34.3% for BIA_{std8}/DXA.

4. Discussion

Body composition measurement could be facilitated by the use of BIA_{std4} or BIA_{std8} in subjects who are able to stand, provided that they are as precise as the BIA_{sup} commonly used for bedridden patients in health care facilities. Our results show that this is indeed the case. Both BIA_{std4} and BIA_{std8} demonstrated intra- and inter-unit precisions almost as good as the BIA_{sup} that we have utilized for 20 years in clinical routine. Thus, they are not only suitable for monitoring the body composition of a subject over time, but their

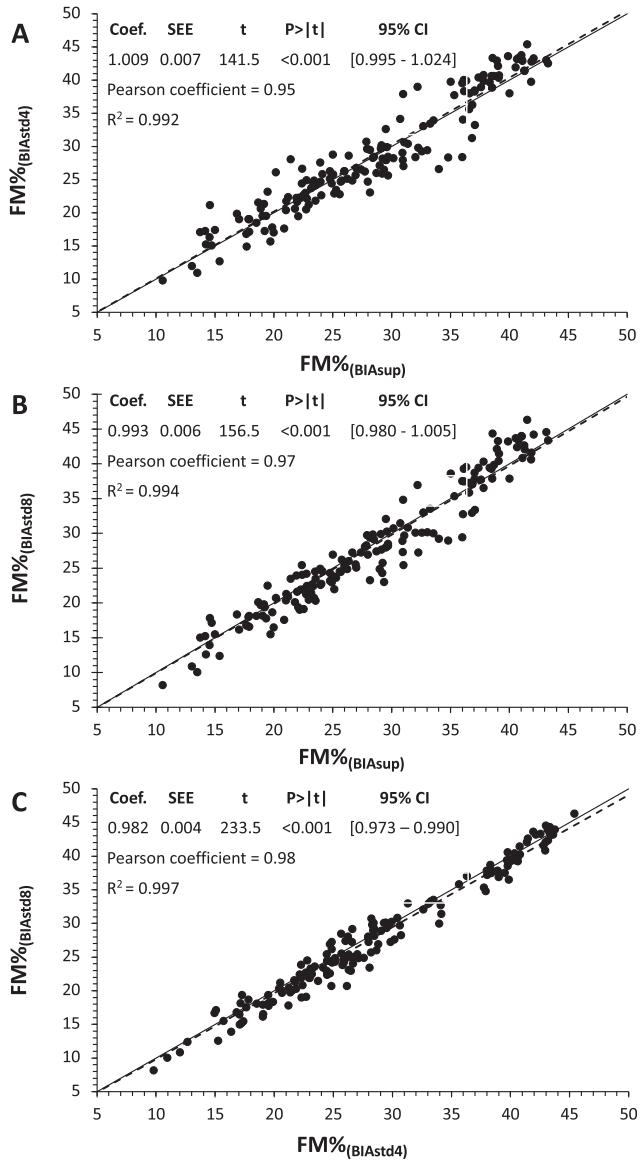

Fig. 2. Scatterplot correlation showing the difference of FM% measurement between BIA_{std4}/BIA_{sup} (A), BIA_{std8}/BIA_{sup} (B), and BIA_{std8}/BIA_{std4} (C). The dashed regression line represents the trend between the FM% measurement mean of both methods.

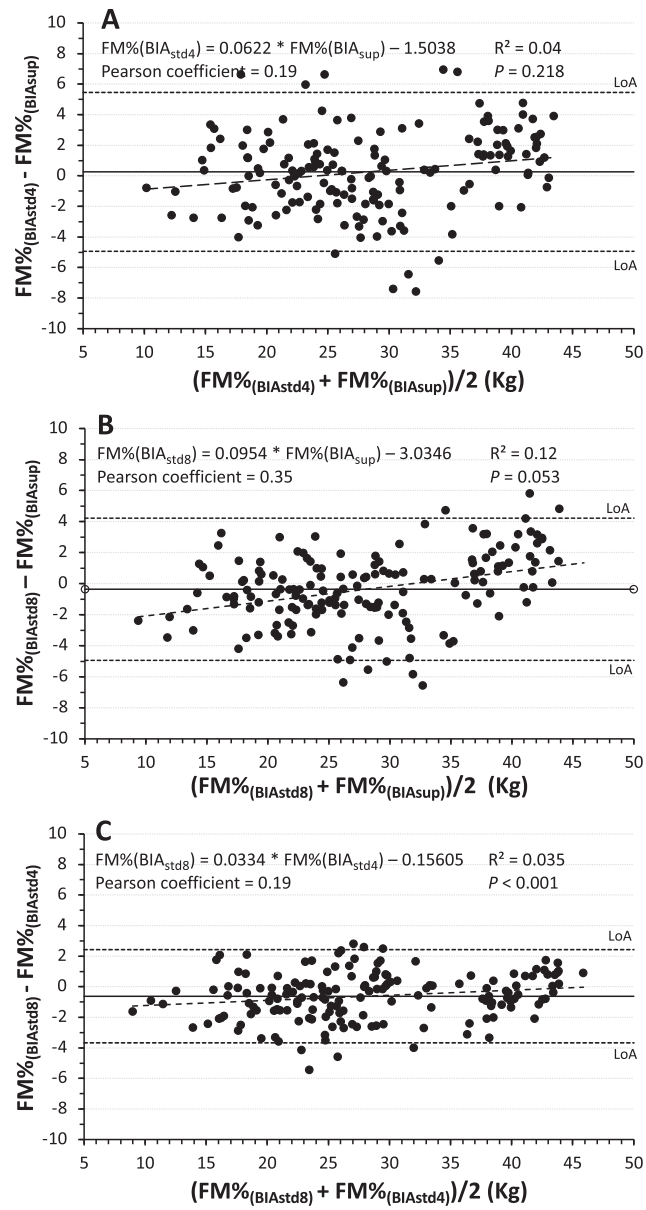

Fig. 3. Adjusted Bland–Altman plots showing the difference of FM% measurement between BIA_{std4}/BIA_{sup} (A), BIA_{std8}/BIA_{sup} (B), and BIA_{std8}/BIA_{std4} (C). The dotted/dashed line represents the constant error or mean bias. The dashed regression line represents the trend between the differences of methods and the mean of both methods.

Table 4
Differences in FM% measurements between BIA_{sup}, BIA_{std4} or BIA_{std8}, in normal weight (N) or overweight (O) men (M) or women (W).

Sex	BMI	Nb	BIA_{sup}[a]	BIA_{std4}[a]	BIA_{std8}[a]
M	N	40	19.2 ± 4.2	18.9 ± 4.0 (−0.3)	18.6 ± 4.7 (−0.6; −0.3)
M	O	40	25.2 ± 4.0	25.3 ± 2.8 (+0.1)	24.6 ± 3.5 (−0.6; −0.6)
W	N	40	29.7 ± 4.5	28.9 ± 3.6 (−0.9)	27.8 ± 4.4 (−2.0***; −1.1†)
W	O	40	38.7 ± 2.7	40.9 ± 2.0 (+2.2***)	40.5 ± 2.9 (+1.8***; −0.4)
M + W	N	80	24.5 ± 6.8	23.9 ± 6.3 (−0.6)	23.2 ± 6.5 (−1.3***; −0.7†)
M + W	O	80	32.0 ± 7.6	33.1 ± 8.2 (+1.1***)	32.6 ± 8.6 (+0.6*; −0.5)
M + W	N + O	160	28.2 ± 8.1	28.5 ± 8.6 (+0.3)	27.9 ± 8.9 (−0.4; −0.6††)

Statistical significance/BIA_{sup} is reported as: *, $P < 0.05$; ***, $P < 0.001$ (ANOVA + post-hoc Bonferroni).
Statistical significance/BIA_{std4} is reported as: †, $P < 0.05$; ††, $P < 0.01$ (ANOVA + post-hoc Bonferroni).
[a] FM% are expressed as Means ± SD (difference/BIA_{sup}; difference/BIA_{std4}).

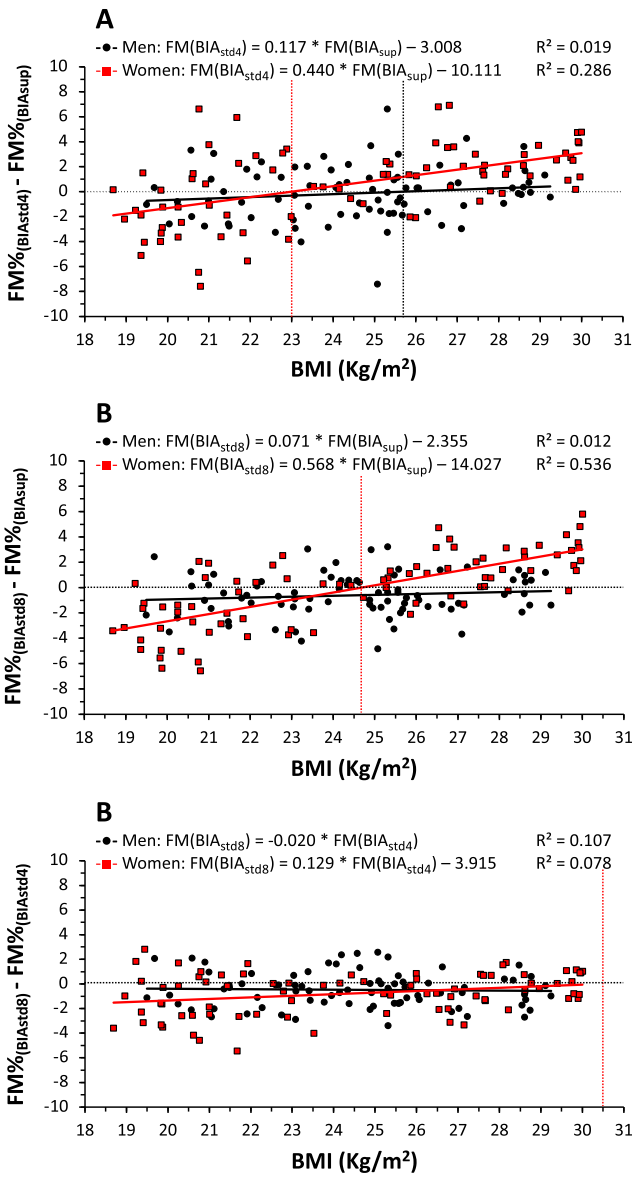

Fig. 4. Adjusted plots showing the difference of FM% measurement between BIA_{std4}/BIA_{sup} (A), BIA_{std8}/BIA_{sup} (B), and BIA_{std8}/BIA_{std4} (C) in function of the sex and BMI. The vertical lines indicate the shift points and the horizontal lines the trend (regression lines) between the two methods. Women = red squares and lines; Men = black point and lines.

units can also be used interchangeably as it is commonly the case in healthcare facilities, without affecting body composition follow-up. Comparison of the different BIA devices using Bland–Altman plots did not show any significant bias between BIA_{std4}, BIA_{std8} and BIA_{sup}. However, the LoAs were quite large, in the order of ±5%, which corresponds approximately to a difference of 1 kg of FM/FFM. Overall, BIA_{std4} and BIA_{std8} tended to underestimate low FM% and overestimate high FM%, compared to both BIA_{sup} and DXA. This effect was more pronounced in women than in men, probably because women have on average larger FM% than men. Consequently, these three devices should not be used interchangeably during body composition monitoring, because BIA_{std4} and BIA_{std8} would tend to accentuate FM% changes over time compared to BIA_{sup} and DXA. Carrion et al. came to the same conclusion while comparing three different BIA devices with arm-to-arm, leg-to-leg and octopolar arrangements [22]. BIA_{std8} was expected to predict changes in body composition similar to DXA, as already shown with another BIA device with foot-to-foot plus hand-to-hand arrangement [23]. However, the tetrapolar arrangement of BIA_{std8} did not provide greater accuracy compared to the bipolar arrangement of BIA_{std4}. It was rather BIA_{std4} which produced the results closest to BIA_{sup} and DXA. Xu et al. also showed that the use of a BIA device with a handle and 8 electrodes did not increase accuracy compared to BIA device with 4 electrodes [24]. These differences in BIA measurement can be explained by different fluid distribution according to the body position. BIA devices in standing position tend to measure lower R and Z values than BIA devices in supine position because of hydrostatic pressure reducing venous return and promoting fluid shifts from the blood vessels to the interstitium [19,25,26]. FM% underestimation was indeed observed with BIA devices in standing position compared to DXA [27]. Inaccuracy in subjects with low FM% may also be explained by the difficulty of whole-body BIAs in assessing visceral fat compared to DXA [28,29]. Another reason may be the difference in conduction of the electrodes depending on their type and placement on the body [19,30]. For BIA_{sup}, the self-adhesive disposable electrodes were placed on the wrists and ankles, whereas for BIA_{std4} and BIA_{std8}, the stainless plate electrodes were in contact with the palms of the hands and the soles of the feet. Therefore, the raw data (R, Xc, Z and ϕ) of these different BIA devices are not directly comparable and must be translated into FFM and FM using an equation/algorithm that is unique to each BIA device. The differences in FM% measurement between these BIA devices are therefore to be mainly attributed to the equations/algorithms which have undoubtedly been validated in different populations with different criterion methods [31]. Regarding BIA_{std4} and BIA_{std8}, the manufacturer's origin (Asia) suggests that the algorithm may not be reliably applied to

Table 5
Difference of FM% measurement between DXA and BIA_{sup}, BIA_{std4} or BIA_{std8}, in normal weight (N) or overweight (O) men (M) or women (W).

Sex	BMI	Nb	DXA	BIA_{sup}[a]	BIA_{std4}[a]	BIA_{std8}[a]
M	N	4	21.0 ± 3.5	15.4 ± 2.3 (−5.6***)	18.3 ± 2.1 (−2.7**)	16.4 ± 2.4 (−4.5***)
M	O	4	23.3 ± 4.8	24.2 ± 6.8 (+0.9)	25.9 ± 3.3 (+2.6)	24.9 ± 5.1 (+1.6)
W	N	4	33.2 ± 6.2	33.2 ± 5.5 (0)	30.6 ± 2.6 (−2.6)	30.4 ± 3.0 (−2.8)
W	O	4	40.3 ± 3.0	39.5 ± 1.9 (−0.7)	42.5 ± 1.6 (+2.2*)	42.9 ± 2.8 (+2.6*)
M + W	N	8	27.1 ± 8.0	24.3 ± 10.3 (−2.8)	24.5 ± 6.9 (−2.6)	23.4 ± 7.9 (−3.7*)
M + W	O	8	31.8 ± 9.8	31.9 ± 9.4 (−0.1)	34.2 ± 9.2 (+2.4*)	33.9 ± 10.3 (+2.1*)
M + W	N + O	16	29.4 ± 9.0	28.1 ± 10.3 (−1.3)	29.3 ± 9.3 (−0.1)	28.6 ± 10.4 (−0.8)

Statistical significance/DXA is reported as: *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$ (ANOVA + post-hoc Bonferroni).
[a] FM% are expressed as means ± SD (difference/DXA).

Caucasian populations. It has indeed been shown that FM distribution and skin texture can differ according to ethnicity [1,32].

4.1. Limitations

The accuracy of DXA to measure FM% can be discussed although it is considered as a criterion method for FM estimation [33]. Indeed, DXA has been shown to overestimate FM% up to 3.7% when compared to other criterion methods, such as hydrostatic weighing and displacement plethysmography [34]. In fact, none of the criterion methods can be assumed to be the "gold standard" as it is the case for the Brozek's reference body obtained from cadaver analysis [35], since the margin of errors between direct and indirect methods are almost equal [36]. This is the reason why we considered it sufficient to compare BIA with DXA only in a subgroup of subjects, provided that they are representative of all BMI and sex categories. Although on a limited number of subjects, our results are consistent with the literature [37].

Another limitation of this study is its restriction to healthy normal or overweight subjects. However, BIA measurements, whatever the type of device, are difficult to interpret in underweight and obese subjects [38], as well as in the setting of some diseases, due to unphysiological water and electrolyte distribution between the extra- and intracellular compartment [2].

5. Conclusion

Both BIA_{std8} and BIA_{std4} show a good intra- and inter-unit precision close to BIA_{sup}, thus making them suitable for rapid body composition assessment, without the cumbersome requirement of single-use electrodes and operator intervention. However, these three devices should not be used interchangeably, because BIA_{std4} and BIA_{std8} tend to accentuate FM% changes during body composition monitoring compared to BIA_{sup} and DXA.

Funding statement

This work was funded by the Foundation Nutrition 2000 Plus (Switzerland).

Author contribution

CP and YD designed the study; KHJ recruited the participants and collected the data; YD analyzed the data and performed statistical analysis; KHJ and YD wrote the paper; All of the authors participated in the critical revision of the manuscript. All authors gave final approval of the version to be submitted.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgment

The authors gratefully thank Julio Conicella and Jérome Lang for performing the DXA examination.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2021.08.010.

References

[1] Kyle UG, Bosaeus I, De Lorenzo AD, Deurenberg P, Elia M, Gómez JM, et al. Bioelectrical impedance analysis - Part II: utilization in clinical practice. Clin Nutr 2004;23:1430–53. https://doi.org/10.1016/j.clnu.2004.09.012.
[2] Haverkort EB, Reijven P, Binnekade JM, De Van Der Schueren M, Earthman CP, Gouma DJ, et al. Bioelectrical impedance analysis to estimate body composition in surgical and oncological patients: a systematic review. Eur J Clin Nutr 2015;69:3–13. https://doi.org/10.1038/ejcn.2014.203.
[3] Fosbøl MØ, Zerahn B. Contemporary methods of body composition measurement. Clin Physiol Funct Imag 2015;35:81–97. https://doi.org/10.1111/cpf.12152.
[4] Kyle UG, Bosaeus I, De Lorenzo AD, Deurenberg P, Elia M, Gómez JM, et al. Bioelectrical impedance analysis - Part I: review of principles and methods. Clin Nutr 2004;23:1226–43. https://doi.org/10.1016/j.clnu.2004.06.004.
[5] Kyle UG, Schutz Y, Dupertuis YM, Pichard C. Body composition interpretation: contributions of the fat-free mass index and the body fat mass index. Nutrition 2003;19. https://doi.org/10.1016/S0899-9007(03)00061-3.
[6] Ceniccola GD, Castro MG, Piovacari SMF, Horie LM, Corrêa FG, Barrere APN, et al. Current technologies in body composition assessment: advantages and disadvantages. Nutrition 2019;62:25–31. https://doi.org/10.1016/j.nut.2018.11.028.
[7] Marra M, Sammarco R, De Lorenzo A, Iellamo F, Siervo M, Pietrobelli A, et al. Assessment of body composition in health and disease using bioelectrical impedance analysis (BIA) and dual energy X-ray absorptiometry (DXA): a critical overview. 2019. https://doi.org/10.1155/2019/3548284.
[8] Genton L, Karsegard VL, Kyle UG, Hans DB, Michel JP, Pichard C. Comparison of four bioelectrical impedance analysis formulas in healthy elderly subjects. Gerontology 2001;47:315–23. https://doi.org/10.1159/000052821.
[9] Kyle UG, Genton L, Karsegard L, Slosman DO, Pichard C. Single prediction equation for bioelectrical impedance analysis in adults aged 20-94 years. Nutrition 2001;17:248–53. https://doi.org/10.1016/S0899-9007(00)00553-0.
[10] Thibault R, Makhlouf A-M, Mulliez A, Cristina Gonzalez M, Kekstas G, Kozjek NR, et al. Fat-free mass at admission predicts 28-day mortality in intensive care unit patients: the international prospective observational study Phase Angle Project and Phase Angle Project Investigators. Intensive Care Med 2016;42:1445–53. https://doi.org/10.1007/s00134-016-4468-3.
[11] Genton L, Mareschal J, Karsegard VL, Achamrah N, Delsoglio M, Pichard C, et al. An increase in fat mass index predicts a deterioration of running speed. Nutrients 2019;11. https://doi.org/10.3390/nu11030701.
[12] Herrmann FR, Graf C, Karsegard VL, Mareschal J, Achamrah N, Delsoglio M, et al. Running performance in a timed city run and body composition: a cross-sectional study in more than 3000 runners. Nutrition 2019;61:1–7. https://doi.org/10.1016/j.nut.2018.10.022.
[13] Foster KR, Lukaski HC. Whole-body impedance - what does it measure? Am J Clin Nutr 1996;64. https://doi.org/10.1093/ajcn/64.3.388s. Am J Clin Nutr.
[14] Lu H-K, Chen Y-Y, Yeh C, Chuang C-L, Chiang L-M, Lai C-L, et al. Discrepancies between leg-to-leg bioelectrical impedance analysis and computerized tomography in abdominal visceral fat measurement open. 2017. https://doi.org/10.1038/s41598-017-08991-y. n.d.
[15] Liao YS, Li HC, Lu HK, Lai CL, Wang YS, Hsieh KC. Comparison of bioelectrical impedance analysis and dual energy X-ray absorptiometry for total and segmental bone mineral content with a three-compartment model. Int J Environ Res Publ Health 2020;17:1–13. https://doi.org/10.3390/ijerph17072595.
[16] Bosy-Westphal A, Later W, Hitze B, Sato T, Kossel E, Glüer CC, et al. Accuracy of bioelectrical impedance consumer devices for measurement of body composition in comparison to whole body magnetic resonance imaging and dual X-ray absorptiometry. Obes Facts 2008;1:319–24. https://doi.org/10.1159/000176061.
[17] Earthman CP. Body composition tools for assessment of adult malnutrition at the bedside. J Parenter Enteral Nutr 2015;39:787–822. https://doi.org/10.1177/0148607115595227.
[18] Maw GJ, Mackenzie IL, Taylor NAS. Redistribution of body fluids during postural manipulations. Acta Physiol Scand 1995;155:157–63. https://doi.org/10.1111/j.1748-1716.1995.tb09960.x.
[19] Jensen B, Braun W, Both M, Gallagher D, Clark P, González DL, et al. Configuration of bioelectrical impedance measurements affects results for phase angle. Med Eng Phys 2020;84:10–5. https://doi.org/10.1016/j.medengphy.2020.07.021.
[20] Messina C, Albano D, Gitto S, Tofanelli L, Bazzocchi A, Ulivieri FM, et al. Body composition with dual energy X-ray absorptiometry: from basics to new tools. Quant Imag Med Surg 2020;10:1687–98. https://doi.org/10.21037/QIMS.2020.03.02.
[21] Dittmar M, Reber H. Validation of different bioimpedance analyzers for predicting cell mass against whole-body counting of potassium (40K) as a reference method. Am J Hum Biol 2004;16:697–703. https://doi.org/10.1002/ajhb.20078.
[22] Carrion BM, Wells A, Mayhew JL, Koch AJ. Concordance among bioelectrical impedance analysis measures of percent body fat in athletic young adults. Int J Exerc Sci 2019;12:324–31.
[23] Antonio J, Kenyon M, Ellerbroek A, Carson C, Burgess V, Tyler-Palmer D, et al. Comparison of dual-energy x-ray absorptiometry (DXA) versus a multi-

frequency bioelectrical impedance (InBody 770) device for body composition assessment after a 4-week hypoenergetic diet. J Funct Morphol Kinesiol 2019;4. https://doi.org/10.3390/jfmk4020023.
[24] Xu L, Cheng X, Wang J, Cao Q, Sato T, Wang M, et al. Comparisons of body-composition prediction accuracy: a study of 2 bioelectric impedance consumer devices in healthy Chinese persons using DXA and MRI as criteria methods. J Clin Densitom 2011;14:458–64. https://doi.org/10.1016/j.jocd.2011.04.001.
[25] Lundvall J, Bjerkhoel P, Quittenbaum S, Lindgren P. Rapid plasma volume decline upon quiet standing reflects large filtration capacity in dependent limbs. Acta Physiol Scand 1996;158:161–7. https://doi.org/10.1046/j.1365-201X.1996.521294000.x.
[26] Rush EC, Crowley J, Freitas IF, Luke A. Validity of hand-to-foot measurement of bioimpedance: standing compared with lying position. Obesity 2006;14: 252–7. https://doi.org/10.1038/oby.2006.32.
[27] Chen KT, Chen YY, Wang CW, Chuang CL, Chiang LM, Lai CL, et al. Comparison of standing posture bioelectrical impedance analysis with DXA for body composition in a large, healthy Chinese population. PLoS One 2016;11. https://doi.org/10.1371/journal.pone.0160105.
[28] Park KS, Lee DH, Lee J, Kim YJ, Jung KY, Kim KM, et al. Comparison between two methods of bioelectrical impedance analyses for accuracy in measuring abdominal visceral fat area. J Diabet Complicat 2016;30:343–9. https://doi.org/10.1016/j.jdiacomp.2015.10.014.
[29] Karelis AD, Aubertin-Leheudre M, Duval C, Chamberland G. Validation of a portable bioelectrical impedance analyzer for the assessment of body composition. Appl Physiol Nutr Metabol 2013;38:27–32. https://doi.org/10.1139/apnm-2012-0129.
[30] Dellinger JR, Johnson BA, Benavides ML, Moore ML, Stratton MT, Harty PS, et al. Agreement of bioelectrical resistance, reactance, and phase angle values from supine and standing bioimpedance analyzers. Physiol Meas 2021;42. https://doi.org/10.1088/1361-6579/abe6fa.
[31] Raeder H, Sørlie Kvaerner A, Henriksen C, Florholmen G, Berg Henriksen H, Kjølsrud Bøhn S, et al. Validity of bioelectrical impedance analysis in estimation of fat-free mass in colorectal cancer patients. Clin Nutr 2018;37: 292–300. https://doi.org/10.1016/j.clnu.2016.12.028.
[32] Meyer KA, Friend S, Hannan PJ, Himes JH, Demerath EW, Neumark-Sztainer D. Ethnic variation in body composition assessment in a sample of adolescent girls. Int J Pediatr Obes 2011;6:481–90. https://doi.org/10.3109/17477166.2011.596841.
[33] Sheean P, Cristina Gonzalez M, Prado CM, Mckeever L, Hall AM, Braunschweig CA. American society for parenteral and enteral nutrition clinical guidelines: the validity of body composition assessment in clinical populations preliminary remarks (intent of guidelines) on malnutrition emphasizes the importance of human body composition i. Clin Guidel J Parenter Enter Nutr 2020;44:12–43. https://doi.org/10.1002/jpen.1669.
[34] Moon JR, Eckerson JM, Tobkin SE, Smith AE, Lockwood CM, Walter AA, et al. Estimating body fat in NCAA Division I female athletes: a five-compartment model validation of laboratory methods. Eur J Appl Physiol 2009;105: 119–30. https://doi.org/10.1007/s00421-008-0881-9.
[35] Brožek J, Grande F, Anderson JT, Keys A. Densitometric analysis of body composition: revision of some quantitative assumptions. Ann N Y Acad Sci 1963;110:113–40. https://doi.org/10.1111/j.1749-6632.1963.tb17079.x.
[36] Raimann JG, Zhu F, Wang J, Thijssen S, Kuhlmann MK, Kotanko P, et al. Comparison of fluid volume estimates in chronic hemodialysis patients by bioimpedance, direct isotopic, and dilution methods. Kidney Int 2014;85: 898–908. https://doi.org/10.1038/ki.2013.358.
[37] Andreoli A, Melchiorri G, De Lorenzo A, Caruso I, Sinibaldi Salimei P, Guerrisi M. Bioelectrical impedance measures in different position and vs dual-energy X-ray absorptiometry (DXA). J Sports Med Phys Fit 2002;42: 186–9.
[38] Schiavo L, Pilone V, Tramontano S, Rossetti G, Iannelli A. May bioelectrical impedance analysis method Be used in alternative to the dual-energy X-ray absorptiometry in the assessment of fat mass and fat-free mass in patients with obesity? Pros, cons, and perspectives. Obes Surg 2020. https://doi.org/10.1007/s11695-020-04614-0. n.d.